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Evaluating Liquefied Natural Gas as a Transitional Pathway to Sustainable and Cleaner Energy Systems: A Comprehensive Review

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Abstract: Liquefied natural gas (LNG) has been widely considered a transitional energy carrier owing to its lower combustion-related emissions relative to coal and oil and its compatibility with existing energy infrastructure. In this study, the role of LNG in facilitating the transition towards sustainable and low-carbon energy systems is critically examined from technological, environmental, and supply-chain perspectives. The thermodynamic principles governing phase conversion from natural gas to cryogenic liquid are analysed, and energy penalties associated with liquefaction, storage, regasification, and transportation are systematically evaluated, with particular attention given to boil-off gas (BOG) generation and mitigation strategies. The integration of LNG within evolving energy systems is further assessed, including its capacity to provide dispatchable backup for variable renewable energy sources and to enhance grid reliability during periods of intermittency. A lifecycle-oriented evaluation is conducted to quantify emissions, energy efficiency, and operational losses across production, liquefaction, maritime transport, storage, distribution, and end-use stages. In addition, supply chain management (SCM) considerations, price parity with conventional fuels, and infrastructure adaptability are examined to determine the feasibility of large-scale deployment. Particular emphasis is placed on the heavy-duty transportation sector in India, where LNG is increasingly considered a lower-emission alternative to diesel due to its higher energy density relative to compressed natural gas (CNG) and suitability for long-haul applications. The analysis highlights both opportunities and limitations, including methane slip, upstream fugitive emissions, and capital-intensive liquefaction infrastructure, which may influence the net climate benefit of LNG. The findings indicate that LNG can contribute to short- to medium-term emissions reduction and operational flexibility when deployed alongside renewable energy technologies; however, its long-term sustainability is constrained by its fossil origin and associated lifecycle greenhouse gas (GHG) emissions. Consequently, LNG is best interpreted as a bridging solution that may facilitate energy system decarbonisation while renewable generation, storage technologies, and hydrogen-based fuels continue to mature.

Keywords: Liquefied natural gas; Sustainability; Supply chain management; Production; Storage; Price-parity; Boil-off gas

1 Introduction

Commodity markets have been under unprecedented pressure in recent years due to geopolitical risk, which carries serious ramifications in the real or financial realm. This has resulted in significantly higher prices for commodities shipped from Russia and Ukraine. Globally, the Russian invasion of Ukraine has had bizarre results. The effects of the Russian-Ukrainian War on the world economy are more noticeable than those of previous geopolitical confrontations. The international community observed a number of results. First, a significant impact on the pricing and supply of important commodity markets, such those for gasses and crude oil.

Global energy consumption is expected to rise at a rate of 1.2% annually on average. Currently, 85% of the world's viable energy needs are met by fossil fuels. Despite increased energy efficiency and the expansion of nuclear and renewable energy sources, worldwide CO_2 emissions are predicted to increase by about 30% between 2015 and

2030 [1]. Over 800 million internal combustion engines (ICE) are in use worldwide, and their number is growing at a rate of almost one per second. Automobiles account for more than 20% of pollution emissions. Globally, governments are enacting more stringent laws [2, 3]. The world faces an urgent need to mitigate climate change by plummeting greenhouse gas (GHG) emissions and transitioning toward a carbon-neutral economy. The importance of liquefied natural gas (LNG) in the energy landscape has increased as countries work to fulfil their obligations under the Paris Agreement and reach net-zero emissions by the middle of the century. Because of its low rate, plentiful resources, and smaller carbon impact than coal and oil, LNG, a flexible energy source derived from natural gas, has drawn interest as a possible transition fuel [3, 4]. Because of the dynamic interaction of technology improvements, economic imperatives, social considerations, and regulatory frameworks, the role of LNG in global carbon neutrality initiatives is both promising and challenging. Although LNG has the potential to be a transition fuel for a cleaner energy future, there are market, regulatory, and environmental issues that must be carefully considered [4].

This paper provides a 15-year update on global LNG sustainability research from 2010 to 2025, analysing 500 articles from the Scopus database. After screening, 168 studies (36%) examined LNG's sustainability impacts in detail, mainly focusing on national (48%) and global (40%) levels. Research predominantly centres on LNG industry practices, with limited holistic life cycle assessments covering environmental, economic, and social impacts across the entire value chain. The review highlights gaps and lessons learned, presenting industry best practices and policies to promote the long-term sustainability of LNG and support global sustainable development efforts.

The bibliometric analysis of LNG and its pertinent sustainability studies from 2010 to 2025 is shown in Table 1. Following the reduction in scope, a thorough examination of the 20 reported literature materials was conducted. During this stage, the sources were grouped according to a number of criteria, including author, year of publication and analytic scope.

Table 1. Bibliometric analysis of the liquefied natural gas (LNG) and its relevant sustainability studies between 2010 and 2025

Reference	Author	Year	Research Analysis
[5]	Boodoo	2010	Overview of LNG export and sustainability of a small gas based economy from Trinidad and Tobago.
[6]	Zhang	2010	Analysis of the developing trend of the global natural gas market and LNG project construction in China.
[7]	Shi et al.	2010	Examine key factors contributing to China's LNG industry's rapid growth. Examine China's LNG infrastructure in detail, both now and in the future.
[8]	Hardisty et al.	2011	The environmental and economic sustainability of carbon capture and storage for LNG.
[9]	True	2011	Focus on sustainability in light of shale gas.
[10]	Beckwith	2012	A social impact analysis on the LNG.
[11]	Zhang and Peng	2012	Analysis of the market position of LNG in China based on energy production and consumption.
[12]	Goncalves	2013	Analysis of current natural gas in the US.
[13]	Tzannatos and Nikitakos	2013	Review and analysis of LNG transportation.
[14]	Robinson	2014	Greenhouse gases (GHG) review and reduction from natural gas.
[15]	Ahmad	2014	Review and analysis of LNG fuelled ship.
[16]	Ma et al.	2015	Review standards and regulations of LNG transportation.
[17]	Agbonifo	2016	Review and analysis of LNG sector.
[18]	Schonsteiner et al.	2016	Life cycle assessment (LCA) study for LNG fuelled ship.
[19]	To and Lee	2017	LCA and GHG of energy sector.
[20]	Köppel et al.	2017	Review and analysis of LNG fuelled vehicles.
[21]	Yang	2018	Environmental impact assessment of LNG industry.
[22]	Gounni et al.	2019	Simulation of urban mobility—LNG fuelled vehicles.
[23]	Al Salmi and Khan	2019	Survey of LNG sector.
[24]	Al-Breiki and Bicer	2020	Mathematical calculation of boil-off gas (BOG) for LNG.
[25]	Boahen and Oppong	2020	Assessment of LNG infrastructure development.
[26]	Al-Haidous et al.	2020	A multi-objective mathematical model for shipping fleet scheduling, routing, and delivery for sustainable LNG supply chains.
[27]	Seithe et al.	2020	A "Well-to-Propeller" LCA of maritime transport fuelled by LNG.
[28]	Arefin et al.	2020	Overview of the LNG as a potential fuel for diesel engines.

Reference	Author	Year	Research Analysis
[29]	Gis et al.	2021	Exhaust emissions of buses LNG and diesel in real drive emissions (RDE) tests.
[30]	Zheng et al.	2022	Evaluation of real-world emissions of China V heavy-duty vehicles fuelled by diesel, compressed natural gas (CNG) and LNG on various road types.
[31]	Adekoya et al.	2023	A comprehensive review of LNG market dynamics.
[32]	Algayyim et al.	2024	Influence of natural gas and hydrogen properties on internal combustion engines (ICE) performance, combustion, and emissions: A review.
[33]	Zhang et al.	2025	Investigation on the emissions of a heavy-duty engine fuelled respectively by CNG and LNG under World Harmonized Transient Cycle (WHTC).
[34]	Mohammadpour and Salehi	2025	A review of alternative liquid fuels in marine engines.
[35]	Abdul-Salam and Abdul-Salam	2025	Reducing carbon emissions in the global energy transition with LNG.

1.1 Research Goals and Structure

The goals of this novel study include the below:

1. Illustrate the preceding studies related to sustainable energy sources, including the LNG production process chain and supply chain.

2. Discuss issues like as boil-off gas (BOG), methane slip, storage and safety, and the potential of LNG supply chain management (SCM) globally.

3. Examine and resolve the gaps in the adoption of LNG as a fuel in India's automobile fleet, the infrastructure for LNG, the obstacles to adaption, favourable pollution rules, and LNG roadmaps.

1.2 Liquefied Natural Gas as a Fuel

Natural gas includes methane and ethane, propane, butane, pentane, etc. Also included are the inert gases viz. nitrogen, carbon dioxide, and water [2]. It is one of the most promising alternative fuels for ICE. It has favourable properties for use in spark ignition engines, like a higher octane number and hydrogen-to-carbon ratio (H/C) ratio, which could provide both reduced operating cost and GHG emissions [36].

Additionally, it can be created from regenerative sources and is readily available in huge amounts from fossil sources. LNG and compressed natural gas (CNG) are the two forms of natural gas now utilized in automobiles. Natural gas is compressed to less than 1% of its volume at typical atmospheric pressure to create CNG. Natural gas is purified and supercooled to -260 °F or -162 ℃ to create LNG. Natural gas is chilled below its boiling point using a process called liquefaction, which eliminates the majority of the fuel's unnecessary components. Methane makes up the majority of the residual natural gas, with trace amounts of other hydrocarbons [37]. While CNG is used for passenger car applications, LNG, which contains more than 98% methane, the cleanest type of natural gas, is a favourable alternative fuel for heavy-duty trucks traveling over long distances [38]. As LNG requires up to 600 times less space than gaseous natural gas, it is useful for tank storage in places without gas pipelines [39]. Natural gas may be transported over oceans and long distances from the nations where it is generated to those where it is needed by converting it to LNG. Natural gas is used for heating and cooking in households, as well as in public buildings, industry, agriculture, and the production of electricity [40].

Recent years have seen a notable increase in the use of LNG due to a number of causes, including rising energy demand, technological advancements, and the growth of LNG infrastructure worldwide [4].

In 1845, British physicist Michael Faraday made the first demonstration of natural gas liquefaction. Engineer Karl Von Linde created the first useful compressor refrigeration device for natural gas in Germany in 1873. By the middle of the 20th century, LNG plants were being used commercially to store natural gas above ground. The Methane Pioneer, a modified World War II Liberty freighter, became the first LNG tanker in history when it successfully transported an LNG cargo from Lake Charles, Louisiana, to Canvey Island, UK, in 1959 [41].

The LNG sector has changed in reaction to shifting economic, technological, and geopolitical situations from its early days of technological innovation to its current position as a major player in the global energy market. LNG's place in the world's energy mix is probably going to increase even more as people look for greener and more effective energy sources [42, 43].

1.3 Liquefied Natural Gas's Potential in the Automotive Industry

LNG is mainly used for transportation and electricity creation. Compared to CNG, LNG offers advantages, such as easier transportation, storage, and safety [44].

Nowadays, LNG is becoming a more and more popular fuel in areas such as:

- Gas supply to small villages and places that cannot be connected to the gas piping,
- Giving gas to customers who are momentarily cut off from the gas pipeline (for example, due to maintenance or repairs), meeting short-term, culminating gas demands during one to four months of the year (i.e., the winter period),
- Supplying final LNG customers as a substitute for gas transported via traditional pipelines. This gas delivery diversification strategy, which supplies the LNG vehicle filling stations, ensures energy independence. The cryogenic tank and the vaporizing installation make up the local LNG supply station [45].

Key challenges for LNG in terms of achieving a global market segment, however, are the absence of infrastructure, harmonised standards and regulations, adoption by automobile manufacturers, and a retrofit system [44].

Many outstanding studies have been conducted in the field of LNG-fuelled engines. LNG is utilized as a monofuel for heavy-duty spark ignition (SI) engines, as a dual-fuel technology in diesel engines, and as a bi-fuel in SI engines with gasoline [33, 46]. A statistical analysis was conducted on over 170 publications concerning various types of LNG-fuelled engines. Table 2 shows an excerpt from this analysis concerning engine power, fuel economy, and controlled exhaust emissions.

Table 2. Statistics on the effect of liquefied natural gas (LNG) fuelled engines

Parameters	% of Total Reference Number			Total No. of References
	Increase	Decrease	Similar	
Power and torque performance	52.9	17.6	29.4	17
Fuel economy	60.9	30.4	8.7	23
CO	3.0	87.9	9.1	33
CO_2	20.0	46.7	33.3	15
CH_4	54.8	25.8	19.4	31
NO_x	66.7	27.3	6.1	33
Particulate matter (PM)	0.0	94.7	5.3	19

The following Table 3 shows the comparison of emissions outcomes between diesel and natural gas (LNG and CNG) operated vehicles:

Table 3. Various emissions comparison between diesel and liquefied natural gas (LNG)/compressed natural gas (CNG) vehicles

Emission Parameters	Diesel	LNG/CNG	Trend w.r.t. Diesel
CO (mg/kWh)	862–1200	650–800	↓
CH_4 (mg/kWh)	5–25	70–220	↑
NO_x (mg/kWh)	200–1800	380–2200	↑
CO_2 (mg/kWh)	770–1203	655–1150	↓
Particulate matter (PM) (mg/kWh)	12.5	7.3	↓

1.4 Advantages and Disadvantages of Liquefied Natural Gas Fuel

1.4.1 Advantages or pros

- High density of energy in liquid form. This means that an LNG-powered vehicle may go up to 2.4 times the distance of its CNG counterpart for a given fuel tank capacity, eliminating the need to stop frequently for refuelling [1].
- The relatively high knock resistance of LNG makes it a good fuel for spark-ignited and dual-fuel engines.
- Reduction in methane slip due to more complete combustion.
- Reducing or eliminating the quantity of pilot diesel using a renewable additive in Dual-fuel mode.
- Produces substantially less soot (black carbon) than diesel.
- Maximum enhancement of efficiency [47]

- Engines that run on LNG are quieter than those on diesel.
- Improvement of local air quality and potential for CO_2 reduction [48].
- LNG offers a higher thermal efficiency and lower specific energy consumption than gasoline [38].
- Using LNG-powered engines to lower particulate matter (PM) and NO_x emissions [49].

1.4.2 Disadvantages or cons

- Requires specialized transportation and storage infrastructure, such as LNG terminals and cryogenic tanks, which can be costly to construct and maintain [47].
- The poor in-cylinder flow conditions and piston shapes, which are attributable to the structure of diesel prototypes and result in low flame propagation rates and slow autoignition, may be the cause of the low thermal efficiency limited by knocking in LNG engines evolved from diesel applications [36, 50].
- Ageing (the evaporation of lighter components) causes changes in LNG's composition over time. Because heavier components have a lower knocking resistance, LNG often has a higher knocking intensity than CNG [38].
- Due to the high ignition energy and low flame speed of the natural gas, a high air-fuel ratio may lead to misfire, high exhaust temperature and an increase in the H/C and CO emissions [51].
- LNG's related safety issues, such as flammability, explosions, and cryogenic burns, are among its main disadvantages.
- A portion of the LNG in the cryogenic tank constantly evaporates as a result of heat entering the tank during storage, transportation, and loading/unloading activities, producing a gas known as BOG.

1.5 Fuel Specifications

LNG is a clear, odourless, non-toxic, non-corrosive, cryogenic liquid at atmospheric pressure [1].

The characteristics of LNG are shown in Tables 4, 5, 6, and 7. The methane number (MN), which is defined as the methane content in a mixture of methane and hydrogen that has the same knocking properties and is examined in the test engine under specified operating conditions, characterizes the knocking properties of a gaseous fuel. A service methane number (SMN) is the outcome of measuring an MN on any engine [38]. Industry guidelines recommend maintaining it above 80 for optimal engine performance using LNG as fuel [52]. The autoignition temperature for the LNG is significantly higher as compared to the other spark-ignited fuel, which overcomes its knocking characteristics [40]. The autoignition temperature is higher for LNG as compared to the other fuel and is shown in Table 8.

Table 4. Liquefied natural gas (LNG) composition (mole%) [47]

Sl. No.	Component	Mole%
1	Nitrogen (N2)	0.185
2	Methane (C1)	97.32
3	Ethane (C2)	1.61
4	Propane (C3)	0.65
5	Isobutane (IC4)	0.115
6	N-butane (NC4)	0.115
7	Isopentane (IC5)	0.005
8	N-pentane (NC6)	0.005

Table 5. Properties of liquefied natural gas (LNG) compared to diesel, gasoline and liquefied petroleum gas (LPG) [40, 53]

Property	LNG	Diesel	Gasoline	LPG
Carbon-to-hydrogen ratio (C/H)	0.25	0.56	0.54	0.38–0.4
Boiling point at 1 bar (°C)	-162	187–343	125	-42
Liquid density (kg/m^3)	423 (-162 °C)	810–890 (15 °C)	740 (15 °C)	573–585 (15 °C)
Theoretical air-fuel ratio	17.2:1	14.3:1	14.6:1	14.7:1
Flammability range in air (% vol)	5–15	–	1.3–6	2.1–9.5
Research octane number	120	–	90–98	100
Motor octane number	130	23–30	80–99	100–110
Cetane number	0	40–60	27	–
Combustion speed (cm/s)	30–35	–	23–27	23–27
Lower heating value (MJ/kg)	50	45	44.4	46

Table 6. Comparison of hazardous properties of liquefied natural gas (LNG) with other fuels [40]

Property	LNG	Diesel	Gasoline	Liquefied Petroleum Gas (LPG)
Toxic	No	Yes	Yes	No
Carcinogenic	No	Yes	Yes	No
Flammable vapours	Yes	Yes	Yes	Yes
Forms vapour clouds	Yes	No	Yes	Yes
Asphyxiant	Yes, but in vapour form	Yes	Yes	Yes, but in vapour form
Other health hazards	None	Narcosis, nausea, eye irritant, others	Narcosis, nausea, eye irritant, others	None
Behaviour if spilled	Evaporates, forming vapour clouds. Portions of cloud could be flammable or explosive under certain conditions.	Evaporates, forms a flammable pool, and environmental clean-up is required.	Evaporates, forms a flammable pool, and environmental clean-up is required.	Evaporates, creating clouds of vapour that, under some circumstances, could be explosive or flammable.

Table 7. Classification of liquefied natural gas (LNG) by density [54]

Composition (%)	LNG Light	LNG Medium	LNG Heavy
Methane	98	92	87
Ethane	1.4	6	9.5
Propane	0.4	1	2.5
Butane	0.1	0	0.5
Nitrogen	0.1	1	0.5
Density (kg/m^3)	427.74	445.69	464.83

Table 8. Autoignition temperature of liquid fuels [40]

Fuel	Autoignition Temperature (°F)
Liquefied natural gas (LNG)	1,004
Liquefied petroleum gas (LPG)	850–950
Ethanol	793
Methanol	867
Gasoline	495
Diesel	Approx. 600

2 Production Mechanism

Fossil fuels like natural gas originate from biological materials transformed over millions of years under pressure and heat. To access natural gas, companies drill wells into underground rock layers, a costly and risky process. Exploration begins with purchasing land or offshore areas, followed by advanced seismic imaging to identify promising zones. After obtaining environmental approvals, which can take years, contractors drill exploratory wells. If tests suggest a viable deposit, additional delineation wells are drilled to confirm the size and properties of the reservoir. This entire process involves significant time, investment, and regulatory compliance before potential extraction [55]. Shale and other low-permeability formations have produced an increasing amount of natural gas in recent years. Hydraulic fracturing, or "fracking" technology, is used to enhance permeability and make it possible to collect natural gas from these tight formations.

After extraction, natural gas is separated from liquids like water, hydrocarbons, and crude oil. It then undergoes further processing to meet pipeline standards for water content, dew point, heating value, and hydrogen sulphide levels. Natural gas most likely originated from decomposed organic layers subjected to heat and pressure over millions of years. Fracking, or hydraulic fracturing, is a common method to access this gas. It involves drilling a vertical well to 2,500–3,000 meters, then turning horizontally into shale rock. A perforating gun creates small holes in the casing, allowing gas to flow into the well for collection and further use [56]. Figure 1 depicts a basic diagram of LNG production.

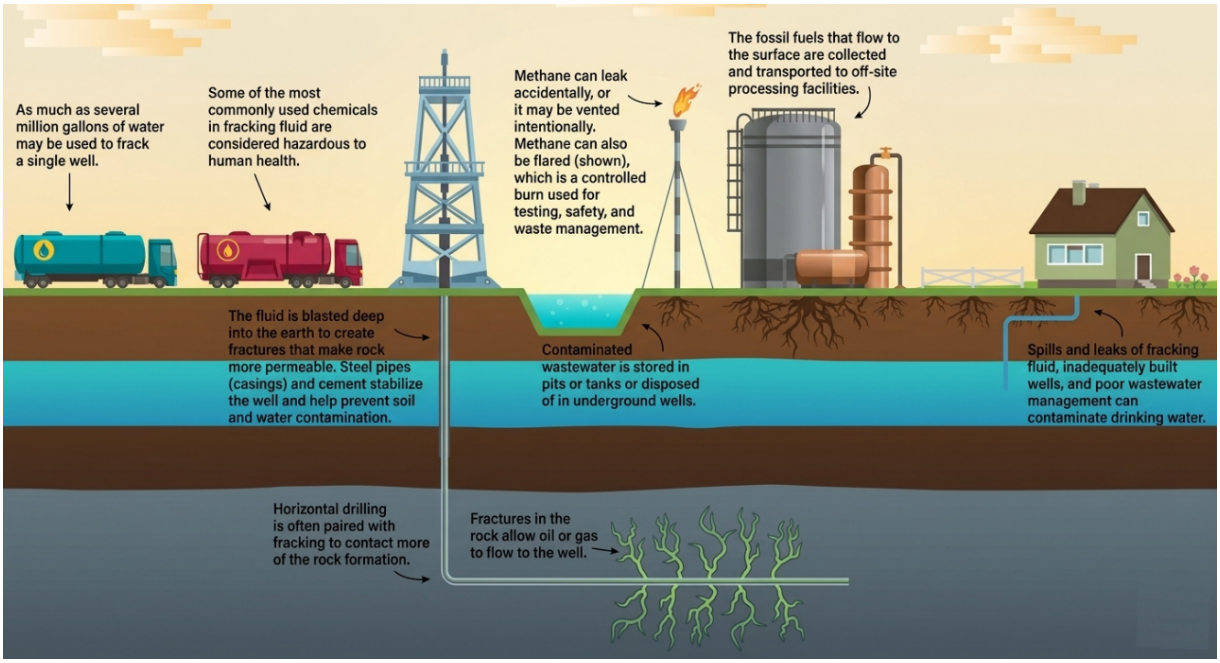


Figure 1. Production and extraction process of liquefied natural gas (LNG) [55]

Fracking begins about three to four months after drilling, using high pressure to fracture the shale rock and release trapped gas and oil. The process involves injecting mostly water (over 90%), along with chemical additives like disinfectants, friction reducers, and acids, to prevent bacterial growth, reduce friction, and dissolve minerals. Sand or clay is added to prop open fissures for continued extraction. Each well uses 3–6 million gallons of water, raising concerns about local water supply impacts. The flow-back water, containing contaminants such as hydrocarbons, salts, heavy metals, and radioactive substances, must be disposed of properly, often at treatment facilities or deep well pits. Recycling is possible but can increase contamination. Despite safety measures like steel and cement casing, mishandling can lead to groundwater pollution, earthquakes, and infrastructure damage. While natural gas emits less CO_2 than coal, methane leaks during fracking pose significant environmental risks, and long-term impacts are still under study.

Fracking has been practiced since the 1940s, but in recent decades it has become increasingly popular. The cost of non-renewable energy is rising as alternative natural gas sources decline, and modern technologies make it extremely accessible. However, due to environmental concerns, many nations and areas have already outlawed hydraulic fracturing [55, 57].

3 Emission Outcomes

Because much of the sulphur that natural gas may contain is removed during the liquefaction process, LNG has a lower density and less sulphur than gasoline. Thus, the amount of sulphurous emissions from re-gasified LNG is negligible [1, 58]. Due to natural gas low carbon-to-hydrogen ratio (C/H), CNG has the lowest CO_2 emissions in relation to the energy content of all fuels. Because there are no nucleation particles during combustion, LNG creates almost no particulate matter. Additionally, because the fuel and oxidizer combine better, it produces fewer gaseous emissions than other fossil fuels, such as CO_2 and NO_x [38]. Also, LNG has a higher methane content compared to CNG. Due to this, a significant reduction in emissions is observed with LNG, i.e., 7.69% in CO, 50% in non-methane hydrocarbon (NMHC), and 6.45% in CH_4 [47]. The overall benefit results in a 40% reduction in emissions and significant minimization of GHG.

Vehicle specific power (VSP) is a function of vehicle speed, acceleration/deceleration, and road slope. It can be shown as kW/t VSP, which is defined as the engine power to counteract vehicle acceleration, wind, rolling, and road slope resistance, is a useful indicator of actual driving emissions. It can be used to compare how various cars' output power and emissions relate to one another. Comparing buses burning different fuels in all VSP levels, LNG buses had a reduction in PM emissions compared to diesel fuel (D100) buses, but NO_x and PN emissions were higher. NO_x and PN emissions of LNG buses were much higher than those of D100 buses [59].

Gis et al. [29] studied exhaust emissions of three city buses in real traffic conditions, one with a SI engine fuelled by LNG, the other with a compressed ignition (CI) engine and a hybrid bus with a CI engine. When compared to a hybrid bus with a diesel engine, the NO_x road emissions from a diesel-powered bus and an LNG-powered bus were

nearly three times lower. The LNG-powered bus's road carbon dioxide emissions were nearly identical to those of a diesel-powered bus, but they were lower than those of a hybrid bus, which released roughly 2% more CO_2. The bus that ran on LNG had the highest CO emissions, which decreased by about 93–94% emissions from NMHC. Comparisons of fossil fuel emissions are shown in Table 9.

Table 9. Fossil fuel emissions comparison(in Pounds per Billion British thermal units of energy input) [29]

Pollutant	Liquefied Natural Gas (LNG)	Oil	Coal
Carbon dioxide	117,000	164,000	208,000
Carbon monoxide	40	33	208
Nitrogen oxides	92	448	457
Sulphur dioxide	1	1,112	2,591
Particulate	7	84	2,774
Mercury	0	0.007	0.016

4 Supply Chain Management

Pipelines are mainly used for short to medium-distance natural gas transport, as they are less expensive than LNG facilities. Pipeline delivery uses 10–15% of the energy compared to about 25% for LNG and produces fewer GHG. However, for longer distances—over 4,800 km on land or 1,600 km offshore—pipelines become less cost-effective than LNG. Both methods have similar energy use and emissions for routes around 13,000 km (pipeline) and 7,500 km (LNG). Disadvantages of pipelines include limited capacity, inflexibility in routing, and dependence on long-term contracts [60]. However, research has revealed that the liquefaction of natural gas to LNG, as well as its storage and regasification, account for the majority of SCM costs (Figure 2).

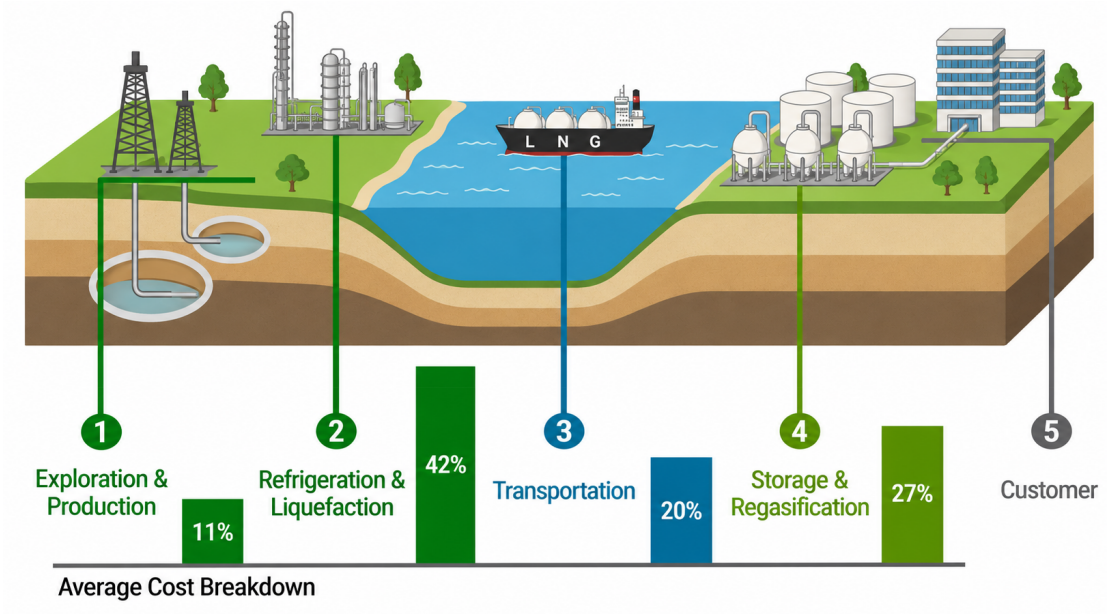


Figure 2. Liquefied natural gas (LNG) value chain and the cost breakdown [61]

LNG is produced in countries like the U.S., Qatar, Australia, and Russia, then transported by truck, train, or ship to nearby markets. It's re-gasified for use in industries, including cooking, heating, and power. LNG also serves as an alternative fuel for heavy-duty transport and shipping [48].

The two basic types of LNG carrier ships have distinctive shapes and are shown in Figure 3. Introduced in 1970, the membrane design tanker has several tanks with a thin (0.5 mm) nickel steel alloy lining that can resist high temperatures. The ship's hull incorporates these tanks. Round containment tanks that rest on the ship's hull and transfer the stress of thermal expansion and contraction onto those supports are a feature of the spherical design tanker that was first deployed in 1971 [55].



Figure 3. Liquefied natural gas (LNG) transport carrier designs

One advantage of LNG is that many regasification facilities can be served by a single liquefaction facility, and vice versa. Moreover, LNG is more flexible than pipeline gas since it can readily change its supply capacity and destination. The ability to tap remote minor gas resources and offshore gas reserves—for which it is not cost-effective to construct a pipeline—is another benefit of LNG [60]. The conceptual energy flow diagram of the LNG value chain is shown in Figure 4.

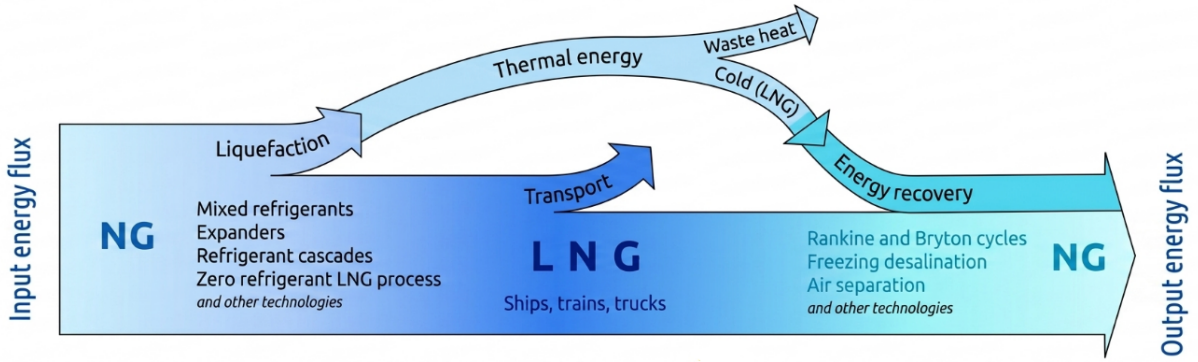


Figure 4. Conceptual energy flow diagram of the liquefied natural gas (LNG) value chain [61]

4.1 Regasification Methodology

The process of turning LNG from a liquid to a gaseous state is known as re-gasification. In order to supply the enthalpy of vaporisation of LNG and heat it from -162 °C to roughly 0–10 °C for pipeline introduction, substantial amounts of thermal energy are needed. LNG regasification is typically carried out at import terminals utilising heat from ambient air, seawater's heat capacity, or partial gas combustion (about 2% of the LNG to be liquefied). Two fundamental benefits of LNG—the comparatively high volumetric energy density (22.2 MJ/L) and physical exergy of the cold gas (1040 kJ/kg at 1 bar pressure)—are not utilised in regasification or pipeline transport. These two advantages make LNG a suitable fuel in mobile applications in waterborne and land transport [62]. The LNG must be re-gasified and heated to room temperature in the special heat exchanger–vaporizer (VAP) before it is sent to the engines. water-glycol (WG) brine, which is used to cool engines, is typically employed as the heating medium for the vaporiser. An LNG pump submerged in the storage tank or a gas compressor downstream in the vaporiser can be used to reach the required gas pressure. A pressure-built unit (PBU), an extra heat exchanger for the evaporation of a tiny amount of LNG, can help create gas pressure inside an LNG tank that is intended to be a pressure vessel [63].

4.2 Cold Energy

The energy extracted from the gases to liquefy them is known as cold energy, and it is just the enthalpy change between storage and ambient temperatures. The thermodynamic availability, which can surpass the cold energy, limits the maximum power and accounts for around 1.7% of the lower heating value of LNG. The cold energy of LNG contains somewhat more latent energy (57%) than sensible energy (43%) [63]. In LNG vaporisers like open rack vaporiser (ORV), submerged combustion vaporiser (SCV), and intermediate fluid vaporiser (IFV), the cold energy of LNG, which is approximately 1040 kJ/kg, is released to the seawater during the regasification stage. LNG then

transforms from liquid (-162 ℃) to gas phase (25 ℃) in the LNG regasification terminals. LNG cold energy will leak into the environment and dissipate into the warming media (such as air and seawater) in the vaporiser units if it is not captured and used [64].

4.3 Liquefied Natural Gas Cold Exergy Recovery Methods

Because it is in cryogenic states, cold energy is a special kind of energy that can be used to provide the thermal energy needed for low-temperature applications [64]. Working fluid loses its low-temperature potential during standard LNG re-gasification, which often uses waste heat or heat from the environment, when its temperature reaches the environmental value. The compressed gas's potential energy is also lost if the pressures are equivalent. The heat required for LNG re-gasification is equal to the amount of cooling power that is available.

The main ways to utilise the potential of LNG cold energy are as follows:

- As a low-temperature heat sink in the Brayton or organic Rankine cycles. Straight cycle converters that reject heat by using the cryogenic working fluid as a low heat source. These systems, like those used in air conditioning and refrigeration, can be efficient, but they are also complicated.
- The primary benefit of direct cooling is its ease of use, as the majority of its parts are readily available. Because the boiling point of LNG is far lower than the temperature at which food is stored, this technique has the drawback of significantly degrading thermal energy. Consequently, there is little energy recovery.
- The thermal compression during LNG re-gasification with further utilisation of high-pressure gas in expansion machines with electric energy generation using thermoelectric generators.
- Conversion to electricity is another way to recover low-temperature exergy. Thermoelectric generators (TEGs) or thermodynamic power cycles can be used to accomplish conversion directly or indirectly. The assumption behind indirect conversion is that the expansion work in a gaseous state at higher temperatures is significantly greater than the compression effort at lower temperatures or in a liquid state. If LNG pressure is raised before regasification and directed toward the expander after regasification, these circumstances can be achieved. We refer to this as the Direct Expansion System [62, 65].

Power generation, air separation, conventional desalination, cryogenic carbon dioxide capture, and natural gas liquids (NGL) recovery are some of the existing LNG cold energy utilisation systems. LNG is mostly utilised in these systems as the heat sink, while it is occasionally utilised as both the heat sink and the feed stream. One of the most popular ways to use LNG cold energy is the cryogenic power-generating cycle. Basic cycles include the Brayton cycle, the organic Rankine cycle, and the direct expansion cycle, where LNG can be used in place of cooling water as the cycle's heat sink. In the distillation column, the air is separated into oxygen, nitrogen, and other elements after being cooled to the cryogenic temperature (below -195 ℃ at atmospheric pressure). Saltwater desalination is an energy-intensive method of removing minerals from saltwater. Because both the desalination plant and the LNG regasification terminal are situated near the shore, it is feasible to use LNG cold energy and lower energy usage. The method of liquefying flue gas to extract CO_2 is known as cryogenic carbon dioxide capture.

Natural gas must be cooled to -30 ℃ or lower in order to use the conventional NGL recovery procedure. Therefore, recovering NGL from LNG during the regasification process is a great way to save energy use and can yield significant revenues. It is highly promising to recover LNG cold energy on the cold chain for data centre cooling, food shipping, and hydrate-based desalination [40]. The LNG low temperature exergy recovery methods overview is shown in Table 10.

Table 10. Liquefied natural gas (LNG) low temperature exergy recovery methods overview [62]

Recovery Method	Applications Applicable for Mobile	Notes
Direct Cooling	Yes	Low efficiency, simplicity, component availability
Direct Expansion	No	Low efficiency, supervision required
Organic Rankine Cycle	No	High efficiency, complexity, supervision required
Brayton Cycle	No	High efficiency, complexity, supervision required
Thermoelectric Generator	Yes	Low efficiency, simplicity, low maintenance, scalability
Absorbed Natural Gas (ANG) Method	Marine: Yes Rail & Road: No	Simplicity, low maintenance

4.4 Delivery and Distribution

Natural aspirated or turbocharged spark ignition engines with basic petrol mixers were utilised in older light and medium-duty LNG vehicles. These engines required fuel delivery pressures of 20 psig or less. However, modern trucks and buses with powerful natural gas engines usually require 75 to 120 psig. Heat transfer during storage and transport causes the pressure to increase to 40 psig by the time LNG is dispensed into the vehicle tank, even though it is delivered to fuelling stations at a saturation pressure of about 10 psig. LNG fuelling stations use techniques including pressurising the storage tank, employing cryogenic pumps, or storing the LNG at a higher pressure by subcooling or adding high-pressure natural gas in order to give the necessary high pressure for contemporary engines [66]. LNG fuel delivery systems are of two types: the vapour collapse (saturated) system and the vapour return system (shown in Figure 5). Only one line hookup from the refilling station is needed for the LNG fuel tank used in the vapour collapse system. The temperature of the residual vapour in the tank can be decreased until the vapour condenses (collapses) into its liquid state by pouring cold LNG into the tank. This lowers the onboard tank's pressure, enabling refuelling. After that, LNG is added to the tank until its operational pressure is reached. In contrast, LNG is directly pumped as a sub-cooled liquid rather than a saturated fluid when using the vapour return system. A second line returns the displaced vapour (Figure 6).

A very significant difference between these two fuel transportation systems is in the way in which the pressures of the onboard fuel tanks are controlled [67].

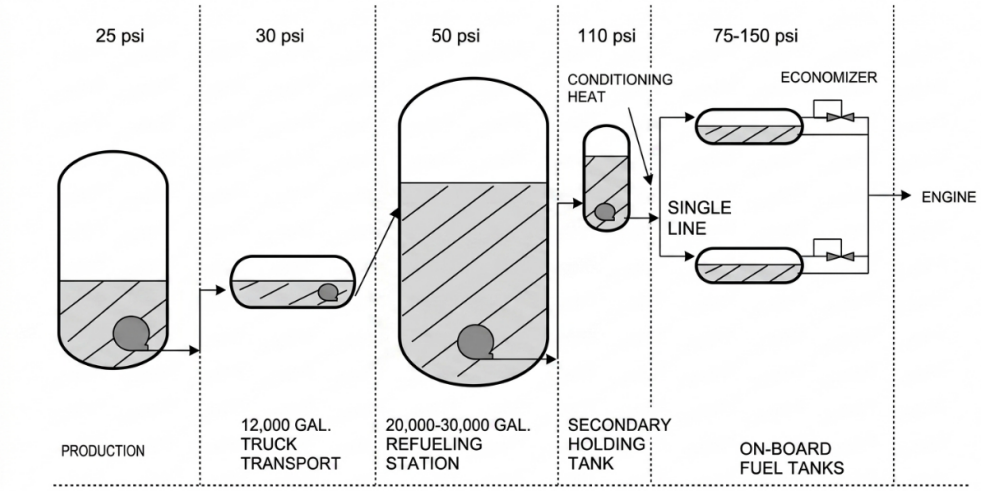


Figure 5. Vapour collapse system [68]

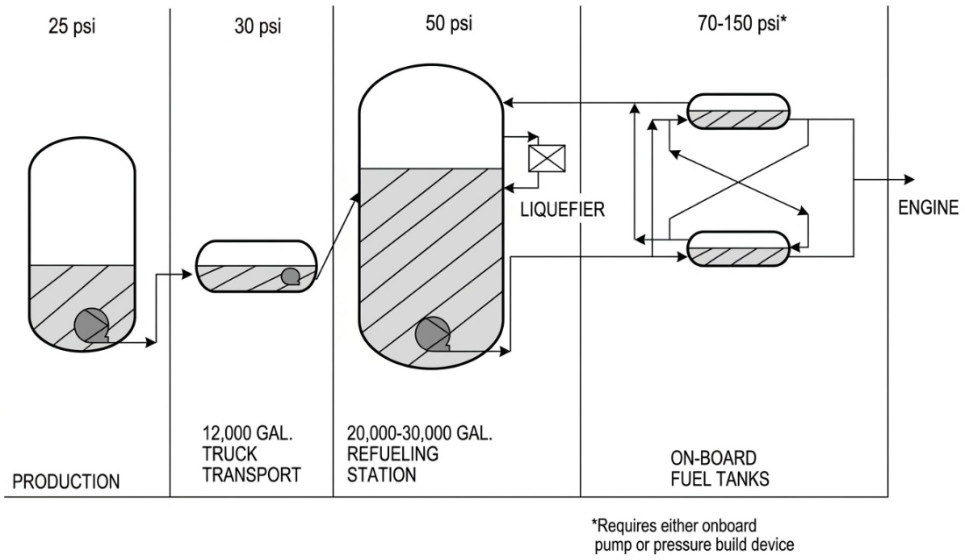


Figure 6. Vapour return system [68]

Table 11 summarizes the benefits and drawbacks of the LNG fuel system, specifically the vapour collapse delivery system and vapour return delivery system.

Table 11. Liquefied natural gas (LNG) fuel transportation systems pros and cons

Vapour Collapse Delivery System	Vapour Return Delivery System
Advantages	**Advantages**
• Single line fill	• Higher onboard fuel density storage
• Absence of pressure build device	• Connectors are at lower pressures
• Greater tolerance for heat leak	• No weathering problems & No atmospheric venting of vapour
	• Uniform pressure drops between tank and engine & No economiser valve
Disadvantages	• Compatible with the second-stage heat exchanger & Has complete refill
• Lower fuel density storage	
• Connectors are at higher pressures	**Disadvantages**
• Susceptible to venting	• Two-line connection
• Susceptible to weathering	• Requires either onsite liquefaction or access to a pipeline
• Requires economiser valve	• Requires an onboard pressure build (conditioning) device or pump
• Can experience large pressure changes between the engine and the tank	• Potential for vapour collapse & Requires a very low heat leak rate tank
• Susceptible to uneven withdrawal from multiple tanks	
• Can experience an incomplete refill	

Refuelling and LNG withdrawal are complicated by multiple tanks, especially for vapor collapse systems, which allow series refilling with overflow to subsequent tanks. Proper station design prevents venting, and vapor can be re-heated, odorized, or returned to tanks for pipeline integration or electricity generation, minimizing GHG emissions [67, 68].

4.5 Liquefied-to-Compressed Natural Gas Filling Stations Concept

A new mixed filling station is proposed alongside conventional CNG and LNG stations, capable of serving all types of natural gas vehicles. It features two dispensers: a gaseous dispenser for CNG and a cryogenic dispenser for LNG, both drawing from a primary LNG container. A 300-bar cryogenic pump supplies liquid LNG to a vaporizer, which compresses it to 200–250 bar for CNG vehicles, requiring less energy than traditional compressor-based stations. Additionally, a withdrawal system transports LNG to the engine via vaporizer and pressure regulator, powering engines with natural gas vapor. Recent innovations include cryogenic injectors that directly pump LNG into combustion chambers [43].

5 Storage and Handling

Natural gas has long been used in transportation; in India, CNG vehicles have been in use for the past 20 years. Although the use of CNG in both private and commercial passenger and goods vehicles has clearly reduced pollution levels, the range of CNG cars is constrained by the amount of gaseous fuel that can be stored on board. Thus, long-haul and larger trucks are the target market for LNG use.

Liquefaction is the most cost-effective way to transport natural gas over long distances and offers a practical and effective way to deliver this valuable energy source to consumers worldwide because it only takes up 1/600 of the volume needed for the same amount of natural gas at room temperature and atmospheric pressure [3, 62].

LNG is stored in double-jacketed cryogenic tanks with an outer carbon steel shell and an inner stainless-steel lining. Insulation is maintained by vacuuming the jackets and filling gaps with perlite. LNG fills no more than 90% of the tank volume, and flexible austenitic steel hoses transfer it from supply sources [39]. The schematic diagram of the LNG Supply Tank is depicted in Figure 7.

Modern LNG marine tankers can carry around 125,000 m^3 across five holds, often with tanks separate from the hull, increasing costs due to special alloys and insulation [43]. These tankers are larger and more expensive than oil tankers, roughly ten times in cost relative to cargo energy. They include safety valves, cryogenic connectors, pressure transmitters, venting systems, and relief valves to manage pressure, prevent explosions, and control boil-off vapours, ensuring safe containment of low-density LNG [45, 69].

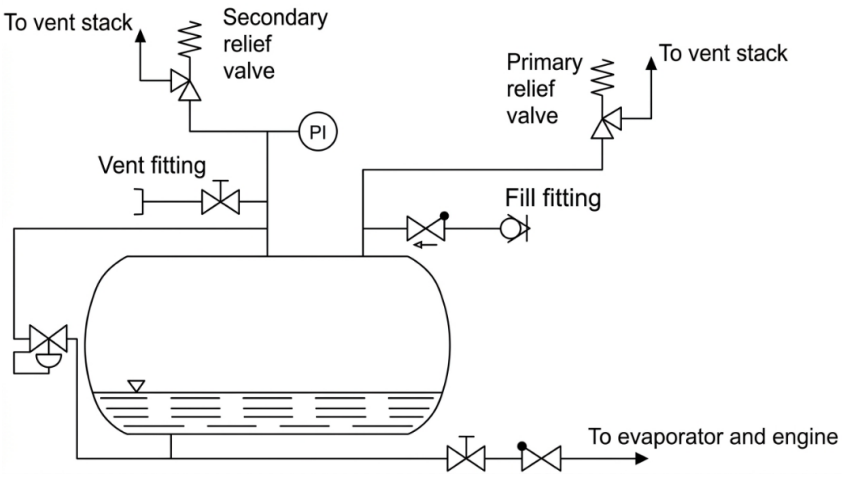


Figure 7. Liquefied natural gas (LNG) supply tank [70]

5.1 Liquefaction Techniques

Gas running through aluminium tube coils is exposed to a compressed hydrocarbon-nitrogen refrigerant in a heat exchanger, which cools the gas throughout the liquefaction process. The LNG is then pumped to an insulated storage tank, where it stays until it can be loaded into a tanker. Heat transfer is achieved as the refrigerant vaporises, cooling the gas in the tubes before it returns to the compressor [55].

The vapour compression refrigeration (VCR) cycle, depicted in Figure 8, is the primary refrigeration system used in the liquefaction of natural gas. Another well-known technology used in this process is the Joule–Thomson valve or turbo-expander [71].

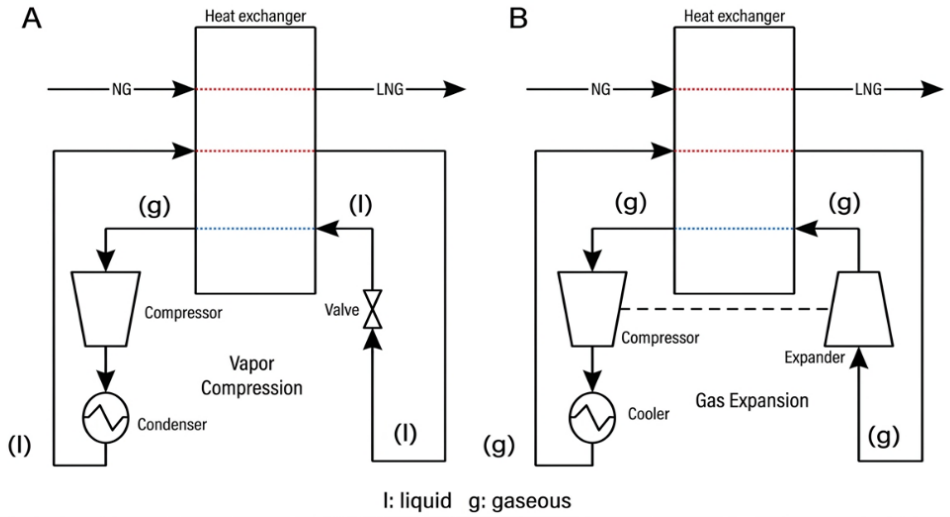


Figure 8. Schematic of vapour compression and gas expansion cycle [62]

These are frequently used principles of natural gas (NG) liquefaction:

- **Cryo-generators** – using the reverse Stirling cycle (among others) with hydrogen or helium to provide cooling power that can achieve temperatures as low as -258 °C.
- **Cryogenic liquids** – nitrogen or oxygen have a boiling point below the boiling point of NG. Technologies for the production of liquid nitrogen are time-proven and available.
- **Cascade cycle** – this technology uses a cascade of heat exchangers, each with a different medium.

Cascade typically uses three refrigeration cycles with pure propane, ethylene, and methane as refrigerant, each at a different temperature. Large LNG plants use this technique the most. There is just one refrigeration cycle in mixed refrigerant technology (MR). A refrigerant made of a blend of light hydrocarbons is needed for this one cycle. The refrigerant in expander-based technology (EXP) is either methane or pure nitrogen. In a single loop, these refrigerants can attain the low temperatures required for NG liquefaction [57, 60].

The intrinsic complexity of each technology—three distinct cycles for Cascade, one cycle with a mixed refrigerant for MR, and one cycle with a pure refrigerant for EXP—is the primary reason for the variations between them (see Figure 9). Cascade has several cooling temperature settings due to the usage of multiple refrigerants. This permits the heat exchangers' hot and cold sides to differ somewhat in temperature.

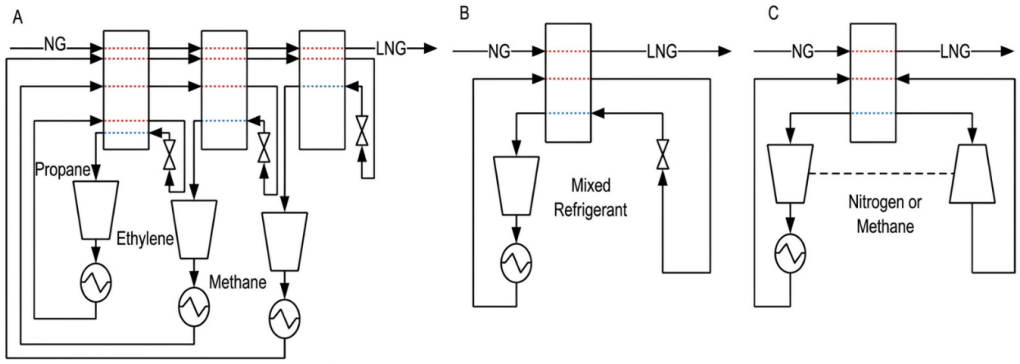


Figure 9. Schematic overview of three liquefaction technologies [60]
Note: A – Cascade, B – Mixed refrigerant technology, C – Expander-based technology

MR uses a refrigerant made of a precisely chosen blend of hydrocarbons to simulate the natural-gas cooling curve. Compared to Cascade, it has an even smaller temperature differential, but it also needs a larger surface area for heat exchange. Throughout the operation, the pure refrigerant in EXP stays in a gaseous state, giving the cooling curve a constant specific-heat value. EXP uses a lot of energy since there is a significant temperature difference between the natural gas and refrigerant, particularly at high temperatures. The heat-exchanger area may be reduced by the significant temperature differential; however, this is offset by nitrogen's significantly lower heat transfer coefficient than hydrocarbons [60].

Utilising an on-site, semi-portable liquefier enables LNG to become a viable, economical, and environmentally clean transportation fuel, as this process incurs no boil-off or atmospheric increases to the greenhouse effect [2]. A comparative study of the evaluation criteria for three LNG Technologies is portrayed in Table 12.

Table 12. Evaluation criteria for three Liquefied natural gas (LNG) technologies [60]

Criteria	Cascade	Mixed Refrigerant Technology (MR)	Expander-Based Technology (EXP)
Application	Onshore large-scale	Onshore, large-scale; small-scale offshore	Onshore small-scale and offshore
Energy efficiency	High	Medium to High	Low
Equipment count	High	Low to Medium	Low
Heat-transfer surface area	Medium	High	Low
Simplicity of operation	Low	Low to Medium	High
Ease of start-up and line-up	Medium	Low	High
Adaptability of feed-gas compositions	High	Medium	High
Sensitivity to ship motion	High	Medium to High	Low
Space requirement	High	Medium	Low
Hydrocarbon-refrigerant storage	High	Medium to High	None
Capital cost	High	Low to Medium	Low

Liquefaction processes use various refrigerants, heat exchangers, compressors, and drivers. The main costly equipment includes compressors, heat exchangers, and drivers. Refrigerants are either mixed or pure; mixed refrigerants replicate natural gas cooling curves, while cascade systems use pure refrigerants with different boiling points. The efficiency order is MR, Cascade, then EXP. Smaller temperature differentials reduce energy use but increase heat exchanger size and capital costs, requiring optimization between refrigerant choice and heat exchange [60].

5.2 Boil-Off Gas Formation

In tanks, LNG is kept and moved as a cryogenic liquid—that is, at a temperature lower than its boiling point. LNG evaporates at temperatures higher than its boiling point, just like any other liquid. A portion of the LNG in the cryogenic tank continuously evaporates due to heat entering the tank during storage, transit, and loading/unloading

activities, producing a gas known as BOG [53]. The design and operation of LNG tanks and ships determine the amount of BOG. The LNG tank experiences an abnormal build-up of pressure due to the rise in BOG. BOG should be regularly removed in order to keep the tank pressure within a safe limit. BOG can be burnt in a gasification unit, re-liquefied, or used as fuel in the LNG supply chain (shown in Figure 10). Additionally, the more volatile elements (methane and nitrogen) boil off first, altering the content and quality of LNG over time. Ageing is the term for this phenomenon.

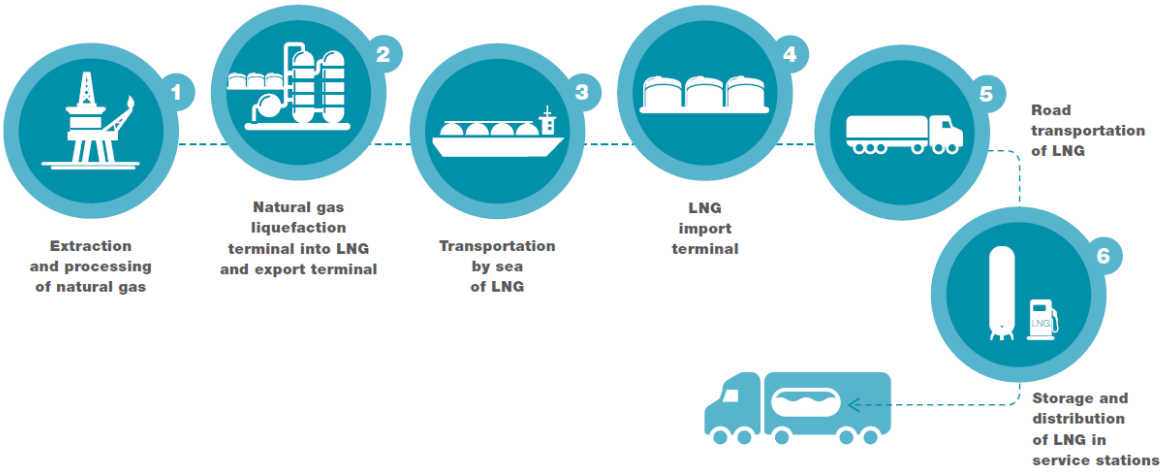


Figure 10. Liquefied natural gas (LNG) supply chain and distribution

The LNG ships themselves produce the majority of BOG in the LNG supply chain. While the ageing process gradually alters the composition, quality, and characteristics of LNG cargo throughout a ship's trip, utilised LNG cargo or losses of LNG cargo due to boil-off reduce the amount of cargo delivered by LNG tankers to the receiving port. Thus, the primary determinants of the economic evaluation of the LNG supply chain are the amount and quality of unloaded LNG [53, 68]. The following are several mechanisms that add heat to the LNG fuel and may cause boil-off:

- Heat leak through the shell of the LNG storage tank
- Natural gas returned from the vehicle tank [53, 67]
- Heat leak through the dispenser
- Heat leak through the fuel hose

The primary cause of LNG boil-off is heat leakage through the storage tank's shell. Additionally, a unique strut design may be employed to reduce the same. To maintain the operating tank pressure within the safe range, BOG should be continuously removed [69].

5.3 Boil-Off Gas Usage Techniques

Ullage vapour in the LNG tank can be used to create energy or liquefied in a liquefier to stop fuel loss in LNG installations. These methods can lower fuel expenses and stop BOG from escaping [68]. BOG is often utilised as fuel in the production process of the liquefaction plant at the loading terminal. Figure 11 depicts a typical block diagram of a liquefaction facility. It is either burnt at receiving terminals or transported via BOG compressors to the re-gasification facility. Depending on the type of propulsion system, BOG may be used as fuel, reliquefied, or burnt in a gasification unit while an LNG tanker is travelling [53]. An independent installation called a re-condensation system (RCS) is used to either reach or increase the LNG fuel tank's holding time. Through insulation and other components of the tank, the RCS's capacity makes up for heat inflow to LNG. The BOG liquefaction process transfers cooling power to the system [62].

The burning of BOG in diesel engines is an efficient technique to use it. However, the quantity of BOG produced during a sea cruise might not be enough to completely satisfy the engine's fuel needs, particularly if there is no cargo load. The idea of dual-fuel engines becomes useful in this situation. Throughout the LNG shipping process, these engines' seamless transition between BOG and conventional diesel fuel ensures a dependable and effective power source [72].

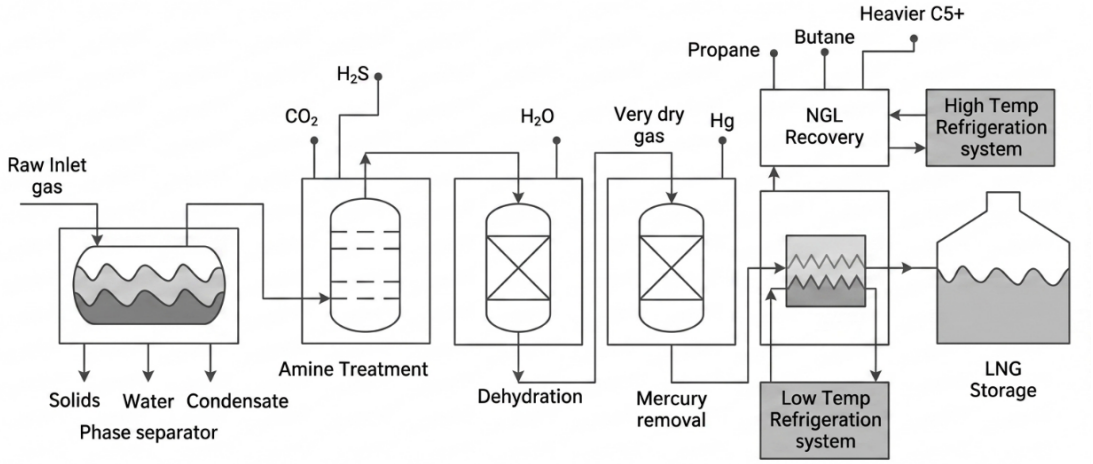


Figure 11. Block diagram of liquefaction plant

5.4 On-Shore Storage

LNG storage facilities vary depending on whether the liquid will be utilised to supply base load gas via long-distance shipping or to meet winter gas shortages (peak shaving facilities to satisfy the seasonal variable gas demand). In the latter scenario, LNG tankers should be used to load and unload entire ship cargoes. In addition to the insulation required to reduce evaporation losses, it is crucial to prevent the LNG cargo from coming into contact with the ship's structure since mild steel becomes brittle below 223 K, which might result in a catastrophic scenario.

If there is enough insulation, evaporation losses for the contents of the tank can be as low as 0.1% per day. Facilities typically accommodate a boil-off of approximately 0.3% for ocean-going vessels. Onshore LNG can be held in double-walled metal tanks that resemble those found in ships, such as aluminium or nickel steel inner vessels or membranes, with external weatherproofing and insulation around them. Furthermore, prestressed concrete tanks can be made underground or built above ground. Lastly, existing subterranean areas that have been specifically designed to store LNG may be utilised. Concrete and natural in-ground tanks have the primary benefit of not requiring containment dykes to collect products from burst or leaking containers. On the other hand, better control over heat escape and the potential for repairs are what make above-ground tanks appealing [1].

5.5 Off-Shore Storage

Due to a shortage of deck area and the ocean environment, small equipment footprints, ease of maintenance, motion sensitivity, and safety are more crucial for an offshore application than efficiency and maximum capacity. To reduce risk, only tried-and-true onshore liquefaction techniques are taken into account for offshore applications. The MR and EXP procedures are more appropriate than the Cascade method due to the features of offshore applications. The EXP method is simpler and requires fewer pieces of equipment than the MR process. Furthermore, the EXP refrigerant does not react to ship vibrations and stays gaseous. Furthermore, nitrogen is safer than MR and does not catch fire. Compared to MR, the EXP process is also easier to operate, faster to start up, and more adaptable to gas composition. The EXP process's high area requirements and poor energy efficiency are its main drawbacks.

Due to the abundance of offshore gas resources, offshore LNG plants are becoming more popular. Due to difficult conditions and limited space, an offshore liquefaction facility is considerably more expensive than an onshore unit. However, due to the gas's low density, the cost of the subsea pipeline, the equipment needed for gas separation, and other factors, moving gas from an offshore extraction platform to an onshore liquefaction plant is equally expensive. The best solution could be LNG floating production, storage, and off-loading (LNG FPSO) and floating LNG (FLNG) [60].

Since the mid-1990s, investors have started projects on the building of FLNG facilities. The company Shell started the operation of the first worldwide FLNG facility in 2011. FLNG technology can unlock offshore NG resources that are technically difficult to extract or too small for building a pipeline connection to shore [58].

6 Challenges with Liquefied Natural Gas Fuel

Environmental issues arise from the extraction, production, and transportation of natural gas as well as the possibility of methane leaks along the supply chain. The main ingredient in natural gas, methane, is a powerful GHG with a 25–40 times greater potential for global warming over a 20-year period than carbon dioxide. Large areas of land must frequently be cleared and natural landscapes altered in order to build LNG facilities, such as liquefaction

plants, regasification terminals, and related infrastructure like pipelines and storage tanks. Particularly in delicate places like wetlands, forests, and coastal zones, this disturbance may result in habitat fragmentation, biodiversity loss, and ecological disruption [73–75].

Geopolitical conflicts, market volatility, financial limitations, public hostility, and competition from renewable energy options are additional difficulties. The social license of LNG projects is also under risk due to worries about methane emissions, water use, and community effects [4].

One of the problems in LNG transportation and storage is the generation of BOG caused by external heat penetration and increased weight due to the required super-insulation tank and pipes, and its combustibility as an extremely low-temperature liquid [52, 53]. Evaporation alters the composition of LNG, which may have an impact on exhaust emissions. There have been reports of serious frost formation issues when LNG is delivered straight to an intake air heat exchanger for a normally aspirated engine [52, 63]. Another issue is the thermodynamic and economic optimisation of the LNG mobile system in terms of holding time and thermodynamic efficiency [62].

6.1 Storage Aspects

LNG boil-off can waste up to 20% of fuel and increase greenhouse emissions during transfer from remote refineries to refilling stations. Heat intrusion from storage, fuelling activities, onboard tank return gas, and seepage through tank shells, hoses, and dispensers causes evaporation. Despite advanced insulation, evaporation persists during transit, but improving insulation reduces cargo volume, causing losses. Re-liquefying BOG is costly and energy-intensive, requiring additional staff. The most practical solution is to use vaporized BOG as fuel for main engines, effectively managing this unavoidable byproduct and minimizing fuel waste and emissions during LNG storage and transportation [68, 72]. LNG tanks let out small amounts of gas as soon as the internal pressure gets too high, creating an atmosphere of flammable gas [76].

6.2 Supply Aspects

When compared to conventional NG transportation, the primary drawback of LNG is that it requires energy to liquefy and then additional heat to re-gasify [65]. Heat leakage causes greater BOG generation than transportation vibrations, although it still makes up more than 10% of all BOG creation and grows quickly when on-road traffic speed increases. In order to accurately account for BOG generations during LNG on-road transportation, the vibration element should not be disregarded, particularly when the LNG tanker is operating at high speed [48]

6.3 Safety Aspects

While LNG is dangerous due to its high temperature, potential for asphyxiation, and fire risk, natural gas is not harmful. Each of these will be addressed in turn. People may sustain frostbite-like cryogenic burns if they come into close touch with the liquid or the substance that contains it. Long-term intake of vapour or cold gas can harm the lungs. Because cryogenic liquids have a low viscosity, they can pass through porous clothing materials more quickly than liquids like water. Additionally, LNG can embrittle materials like rubber and carbon steel, which can lead to cracking failures. As the gas warms up, it turns colourless and odourless, making it undetectable to human senses, even though a release is frequently first visible as a cloud due to the production of frost from the atmosphere. As a result, it is simple to go into an area where the oxygen content is so low that unconsciousness is nearly instantaneous. With a Lower flammable limit (LFL) of 4–5% by volume in air and an upper flammable limit (UFL) of roughly 15%, depending on temperature, it is obviously quite flammable as a fuel gas. A flash fire is the primary risk of an LNG spill into the open space above earth, while a physical explosion could occur above water. Only under extremely rare circumstances can a vapour cloud explosion (VCE) take place in cramped areas (overfilled zones) [39].

The flammable gas vapour that results from spilling LNG on the ground or on water will typically evaporate into the atmosphere and no fire will occur if it does not come into contact with an ignition source (a flame, spark, or source of heat of 540 °C or greater at atmospheric pressure). LNG vapour, or methane, needs the highest temperature for autoignition when compared to other liquid fuels.

6.4 Hazards Associated with Liquefied Natural Gas

Explosion: When a material abruptly changes its chemical state—that is, ignites or is uncontrollably released from a pressurised condition as a result of structural failure—an explosion occurs. Because LNG is kept at a very low temperature—roughly -162 °C—no pressure is needed to keep it liquid; in other words, LNG is kept at atmospheric pressure. Consequently, an explosion will not happen right away if the container cracks or punctures.

Vapour clouds: LNG starts to warm up as it exits a temperature-controlled container, changing the liquid back into a gas. At first, the gas is heavier and colder than the surrounding air. Above the discharged liquid, it produces a fog, a cloud of vapour. The gas interacts with the surrounding air and starts to disperse as it warms. Only when the vapour cloud is concentrated within its flammability range and comes into contact with an ignition source will it catch fire.

Freezing liquid: Direct human contact with the cryogenic liquid will cause the place of contact to freeze if LNG is released. To protect themselves from the freezing liquid, all facility workers are required to wear gloves, face masks, and other protective gear.

Rollover: LNG supplies of different densities do not initially mix when they are placed into a tank one at a time. Rather, they arrange themselves in shaky layers inside the tank. These layers may eventually rollover on their own to stabilise the tank's liquid. The density of the lower LNG layer changes as it is heated by a typical heat leak, eventually becoming lighter than the upper layer. The tank may eventually develop cracks or other structural problems as a result of the excess pressure. Operators unloading an LNG ship measure the density of the cargo and, if needed, modify their unloading operations to avoid stratification. Rollover protection devices for LNG tanks include pump-around mixing systems and distributed temperature sensors.

Rapid phase transition (RPT): Because LNG is less thick than water, it floats and evaporates when released on water. Large amounts of LNG may evaporate excessively quickly on water, resulting in a fast phase transition. Only when there is mixing between the LNG and water can an RPT happen. Small explosions and bursts big enough to possibly harm lightweight structures are examples of RPTs [40].

7 Global Prospects for Liquefied Natural Gas as a Vehicle Fuel

Government assistance through policy initiatives is one of the primary accelerators for the adoption of any alternative fuel vehicle, as can be seen from the review of successful models used globally in nations like China, Italy, Spain, the Netherlands, and Germany [77]. LNG policies are adopted globally and are compared in Table 13.

Table 13. How liquefied natural gas (LNG) policies are adopted globally [77]

Sl. No.	Incentives	China	Italy	Spain	Netherlands	Germany
1	Subsidy on purchase of new LNG heavy duty vehicle (HDV)	✓	✓	✓	✓	✓
2	Monetary benefits for scrapping old diesel HDV	✓	×	×	×	×
3	Lower LNG tax vs diesel	✓	✓	×	✓	✓
4	Toll free exemptions for LNG HDV	×	×	×	×	✓
5	Procurement policy to promote LNG HDV	×	✓	×	×	×
6	Ban on Diesel HDVs in the select regions, where LNG HDV allowed	✓	×	✓	✓	×
7	EFuel Efficiency and/or emission targets for HDVs	✓	✓	✓	✓	✓
8	National targets on LNG HDV and associated infrastructure	✓	✓	✓	✓	×

More than half of all alternative fuels in the world today are dry gas (natural gas and propane), making LNG one of the most significant fuels of the 21st century. By 2030, the demand for LNG is expected to rise to 500 million tonnes, particularly given the state of the world economy [2, 62]. The CY 2024 – LNG import countries vs. million metric tonnes per annum (MMTPA) and percentage is depicted in Figure 12.

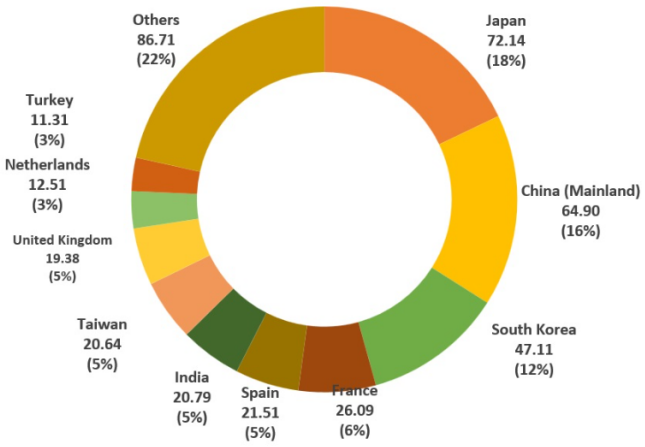


Figure 12. CY 2024 – liquefied natural gas (LNG) import countries vs. million metric tonnes per annum (MMTPA) and percentage [78]

The world has committed to become carbon neutral by the middle of the century as per the 2015 Paris Agreement. According to Institute for Energy Economics and Financial Analysis (IEEFA), the largest number of new LNG plants ever constructed in a single year—roughly 57 MMTPA—will begin operations in 2026, with 44 MMTPA in 2027 and 43 MMTPA in 2028 expansions.

Australia, the world's top LNG exporter in 2021 and 2022, will fall to a distant third place among global providers as the majority of the new LNG capacity to be completed by 2028 is focused in the United States and Qatar. In the meantime, significantly more LNG capacity is being built in African countries, Canada, and Russia. Figure 13 and Figure 14 provide a quick overview of these facts.

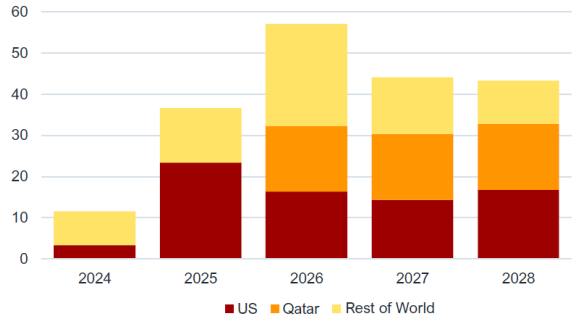


Figure 13. Global liquefied natural gas (LNG) supply additions 2024–2028 million metric tonnes per annum (MMTPA) [79]

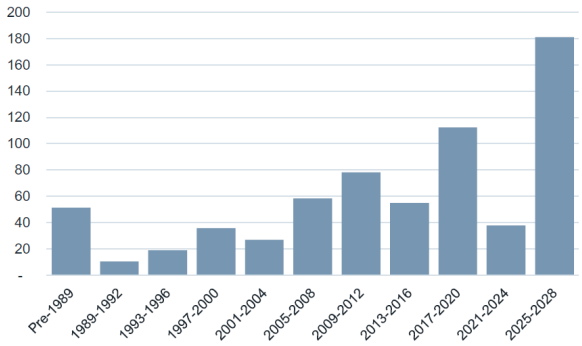


Figure 14. Net global liquefied natural gas (LNG) supply additions million metric tonnes per annum (MMTPA) [79]

Global LNG production capacity is expected to increase, but operational difficulties and diminishing feedstock gas production at many facilities will limit overall output. In recent years, over one-third of the world's LNG plants have experienced serious problems with gas supply and mechanical dependability. The global LNG fleet barely ran at 87% of its rated capacity in 2023 due to these disruptions. Figure 15 displays the global LNG exports as of 2023.

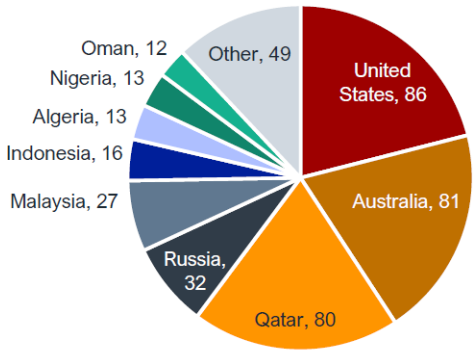


Figure 15. Global liquefied natural gas (LNG) exports, 2023 [79]

Since opening its first LNG facility on the Gulf Coast in 2016, the U.S. has become the world's largest LNG exporter, with seven operational plants and a capacity of 92.3 MMTPA—about 20% of global supply. In January 2024, the U.S. Department of Energy halted new LNG export authorizations to Non-Free Trade Agreement countries to reassess impacts on prices, climate, and communities. U.S. LNG contracts offer buyers significant flexibility, enabling them to resell or cancel shipments for cost reasons. During periods of low prices, such as mid-2020, many LNG plants were underutilized, with cancellations causing reduced output and temporary shutdowns amid oversupply.

Due to the high costs of both building and new gas sources, new Australian LNG projects are typically seen as being globally uncompetitive. Australia's Safeguard Mechanism, which mandates that any new gas fields have zero reservoir CO_2 emissions, presents additional difficulties for the business. Carbon capture and storage (CCS), which has a lengthy history of poor performance and will probably raise the cost of LNG projects, will probably be necessary for the development of these fields.

Qatar has started a significant LNG expansion initiative that would increase its liquefaction capacity by 64 MMTPA over the next six years. Prior to 2020, Qatar was the world's top exporter of LNG; in 2023, it ranked third. The offshore North Field's attractive economics, which generate large amounts of petrol and liquid condensates that are used as feedstock for diesel, jet fuel and petrol, support Qatar's LNG aspirations. Even at low LNG prices that would bankrupt most other producers, Qatari LNG projects are able to turn a profit because of revenues from liquids.

Russia's LNG exports have remained strong, with 32 Mt exported from the nation's four operational plants in 2023, despite a decline in pipeline gas exports following the invasion of neighbouring Ukraine. Despite the European Union's 2022 commitment to stop using Russian fossil fuels by 2027, EU countries have bought more Russian LNG in the past two years than they did prior to the invasion of Ukraine.

India is making investments in new infrastructure for import terminals and pipelines. As of September 2023, 24 MMTPA of new LNG regasification capacity and about 10,009 km of gas pipes were being built. Indian LNG importers seek to expand supplies obtained from long-term contracts due to recent spot market volatility. For instance, Petronet extended a 20-year agreement to purchase 7.5 MMTPA beginning in 2028 in February 2024. Additionally, Gas Authority of India Limited (GAIL)has inked a ten-year deal with Vitol Asia to purchase one MMTPA starting in 2026 [79].

The United States has solidified its position as the world's top exporter by emerging as a global powerhouse in LNG exports. LNG shipments from the United States have increased dramatically in recent years; in 2021, they accounted for around 120 billion cubic meters, or 22% of all LNG trading worldwide [26]. The launch of new liquefaction facilities and the European market's withdrawal from Russian gas supply have been the main drivers of this expansion [80].

Europe has greatly expanded its imports of LNG in reaction to the disruption of Russian pipeline gas, with a 60% increase in 2023. Leading nations that have diversified their energy sources include France, the Netherlands, Spain, and Poland. To improve its capacity to import LNG, the EU has also established new liquefaction facilities in the Netherlands [58].

Although Asia has historically been the biggest market for LNG imports, net imports in the region decreased by 8.7% in 2022. This is mostly caused by a 14% decline in Indian LNG imports as a result of high pricing and a 20% decline in Chinese imports as a result of declining local demand. However, it is anticipated that China's LNG imports would increase in 2023 and may surpass 2021 levels by 2024 [58, 81].

Africa has historically been a small player in the global LNG market, but it has the potential to become a major supplier in the future. Utilising their wealth of natural gas resources, nations like Mozambique, Nigeria, and Equatorial Guinea are creating new LNG export initiatives [82].

Several countries in Latin America, such as Brazil, Chile, and Argentina, have been increasing their LNG imports in recent years to diversify their energy sources and meet growing domestic demand [58]. However, the region's LNG import growth has been uneven, with some countries facing challenges in securing reliable and affordable LNG supplies [80].

The continuous energy shift and regulatory uncertainties provide major obstacles as the global LNG market continues to develop. The future of the LNG sector will be greatly influenced by policies that strike a balance between economic factors, environmental sustainability, and energy security [45, 80].

8 Indian Scenario

8.1 Liquefied Natural Gas Imports

India's LNG imports have grown significantly in recent years, from 31% of the country's natural gas supply in 2012 to more than 50% in 2019 [83]. India imported around 20.79 MMTPA of LNG in 2022, which is 5% of the total world LNG trade [84]. Qatar remains the main supplier of LNG to India due to the relatively short transportation distance. However, due to an increase in domestic natural gas production, India's dependency on imported LNG is expected to drop from 53% in Fiscal Year (FY) 21 to about 45% by FY26. Over the last three years, about 30 million

standard cubic meters per day (MMSCMD) of new production have been added, and an additional 15 MMSCMD are anticipated to be online in FY25 [85].

8.2 Liquefied Natural Gas Infrastructure

India has been expanding its LNG import capacity, which more than doubled during the past 10 years [84]. With a total capacity of roughly 47.7 million tonnes, the nation currently has six LNG import terminals. By 2023, four additional LNG import facilities that are presently under development should be operational, adding 2.5 Bcf/d of LNG import capacity [83]. In order to reach a 15% share of natural gas in the primary energy mix by 2030, the government plans to build regasification capacity of 70 MMTPA by 2030 and 100 MMTPA by 2040. Establishing virtual pipelines and facilitating infrastructure for LNG transit to all areas of consumption are other key objectives of the policy [84].

8.3 Liquefied Natural Gas Consumption

Up to FY20, India's natural gas consumption increased steadily due to the government's campaign for cleaner fuel. By 2030, the nation wants to raise the proportion of natural gas in its primary energy mix from 6% to 15%, concentrating on industries including petrochemicals, power, refineries, fertilisers, and city gas distribution. India is expected to have its highest annual petrol consumption in FY24 due to strong demand across all key segments. In order to achieve India's energy objectives and lessen its need for imported LNG, this trend is essential [85].

8.4 Emission Regulations Favourable to Liquefied Natural Gas

The Indian government has put in place a number of measures to encourage the use of LNG in a number of industries, especially mining and transportation. These regulations seek to boost the use of LNG, guarantee fair distribution, and create a stable and supportive environment for LNG-related operations. The government has pushed for the construction of LNG filling stations along the Golden Quadrilateral in order to initiate the development of an LNG-fuelled transportation ecosystem. By 2024, the initiative aims to have one lakh LNG-powered buses or trucks. The Inland Waterways Authority of India (IWAI) is also working on a number of projects to build LNG infrastructure along the waterways so that LNG can be used as a fuel for transportation.

In order to enhance the dynamics of the gas market and increase consumption, the government has also implemented regulatory changes. These include a new priority system for unregulated domestic gas volumes, modifications to the fertiliser procurement procedure, a new domestic gas pricing formula, and modifications to the gas transmission tariff structure [84].

8.5 Indian Government Notifications (Liquefied Natural Gas Roadmaps)

The Indian government has put in place a number of measures to encourage the use of LNG in a number of industries, especially mining and transportation. These regulations seek to boost the use of LNG, guarantee fair distribution, and create a stable and supportive environment for LNG-related operations [48]. The government has set high goals for the development of LNG infrastructure. In order to reach a 15% share of natural gas in the primary energy mix by 2030, it seeks to develop regasification capacity of 70 MMTPA by 2030 and 100 MMTPA by 2040. Establishing virtual pipelines and facilitating infrastructure for LNG transit to all areas of consumption are other key objectives of the policy [77].

The government has pushed for the construction of LNG filling stations along the Golden Quadrilateral in order to initiate the development of an LNG-fuelled transportation ecosystem. By 2024, the initiative aims to have one lakh LNG-powered buses or trucks. Additionally, the IWAI is working on a number of projects to build LNG infrastructure along waterways so that LNG can be used as a fuel for transportation [86].

The new domestic gas pricing formula, introduced in March 2023, includes a 10% linkage to crude oil and has a ceiling of US$6.50 per MMBtu and a floor of US$4 (increasing by US$0.25 every year post-2025). This formula aims to make gas more affordable for the city gas sector and reduce the cost of subsidies in the fertiliser sector. In addition, the government has altered the gas transmission tariff structure to lower transmission costs for city gas and inland businesses, promoting a move away from fuel oil and coal. It is anticipated that these regulatory adjustments will make LNG more affordable and change India's price sensitivity threshold, enabling more demand growth at higher spot prices [86].

India's LNG market is one of the few growing gas markets globally. The country has long-term LNG contracts of around 20 MMTPA, which constitutes around 95% of the total LNG consumption in 2024 [84].

8.6 Challenges for using Liquefied Natural Gas as a Transportation Fuel in India

Even if the usage of LNG to power automobiles has increased, there are still a number of obstacles that prevent LNG from being used in natural gas vehicles (NGVs). These difficulties are:

1. There are several gaps in international standards as a result of the comparatively small number of NGV laws. One barrier to the manufacture and use of LNG equipment is the absence of international laws and standards.

2. Regarding energy for transportation and the permissible passage of pipeline or maritime routes within each nation's borders, India has inconsistent national interests and policies. This is an issue that restricts the development of various NGV types in nations with robust LNG markets.

3. Heavy duty vehicle (HDV) and machine manufacturers have not yet modified their products to use LNG. There are hardly many LNG-powered HDVs and machineries on the market. This is a factor that restricts the rise in the use of LNG fuel.

4. Because LNG must be available at different pressures, the systems installed in HDVs and machines function differently. This suggests that building LNG fuelling stations is a difficult undertaking.

5. One issue that needs to be addressed is the inconsistent quality of LNG. The uniformity of the process may be impacted by different importers' requests for different product requirements and final LNG product purities.

6. The fuel tank and NG engine are the primary reasons why LNG HDVs are more costly than diesel HDVs; the difference is close to Rs. 12+ lakhs. The cost of retrofitting a single 450L/180 kg tank to convert diesel to LNG is considerably expensive, ranging from Rs. 12 to 16 lakhs.

9 Vision and Mission

India has outlined a plan to raise the proportion of natural gas in the primary energy mix to 15%. The Ministry of Petroleum and Natural Gas (MoPNG) released draft LNG policy 2, which focuses on methods to expand the use of LNG as a fuel for transportation. Along the Golden Quadrilateral, the groundwork for India's first fifty LNG petrol stations was established in order to carry out the same. To boost the use of LNG as a fuel for transportation and in the mining industry, the Indian government has published a draft LNG policy [77].

Under the Paris Agreement, countries have united to confront these issues by promising to keep global warming well below 2 ℃ and to work toward a 1.5 ℃ limit. A major change in energy systems is necessary to meet these lofty objectives, with an emphasis on lowering dependency on fossil fuels and switching to low-carbon and renewable alternatives. LNG has become a viable bridge fuel in this regard, providing a route to decarbonisation. LNG has a lower carbon intensity than coal and oil, which have high carbon contents and produce large amounts of GHG when burnt [4].

It is expected that there will be at least a few dozen LNG-fuelled ships built or modified annually, and hundreds or possibly thousands beyond 2030. The International Marine Organization's (IMO) objectives include that by 2050; all GHG emissions must be at least 50% lower than they were in 2008 [62].

10 Ongoing Development of Internal Combustion Engines for Liquefied Natural Gas Fuel (Engine Performance and Modifications)

The schematic diagram of the working principle of the LNG engine is shown in Figure 16.

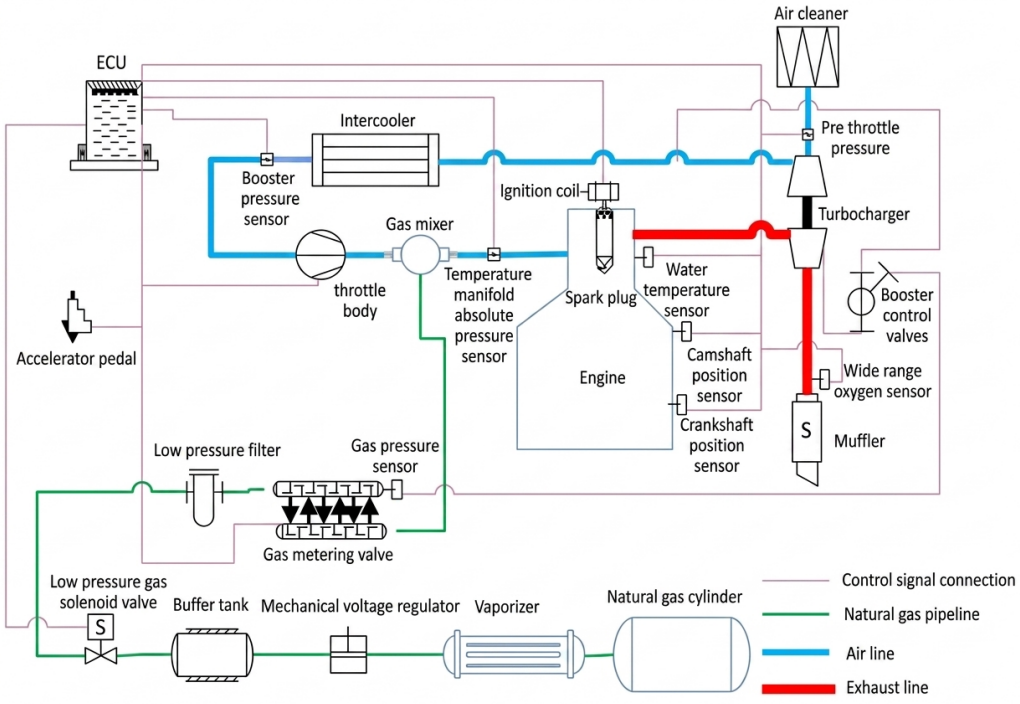


Figure 16. Working principle of liquefied natural gas (LNG) engine [3]

10.1 Engine Performance and Modifications Aspects

ICEs can use LNG as a cleaner alternative fuel, mainly in dual-fuel designs where LNG is used in place of diesel and ignited by a diesel pilot. Due to pollution reductions in the face of stricter restrictions, this strategy has gained popularity in the maritime, heavy-duty vehicle, and power-generating industries. Performance improvements and technological advancements from 2020 to 2026 are highlighted in recent literature, with an emphasis on efficiency and methane slip reduction.

Core Technology: Using a tiny diesel pilot (usually 20–30% of energy) for compression ignition in a two-stage combustion process, dual-fuel LNG-diesel engines combine gaseous natural gas with air. At high loads, this produces torque and brake mean effective pressure (BMEP) comparable to diesel, with up to 78% natural gas substitution at mid-speeds like 1300 rpm. Gaseous fuel displacement causes a modest decrease in volumetric efficiency, whereas lean-burn operation increases thermal efficiency by 5–15%. In order to exceed Euro 7/IMO emissions while reducing costs, recent advancements in LNG fuel systems for heavy-duty trucks have focused on dual-fuel integration, cryogenic tank improvements, and high-pressure direct injection (HPDI). In the face of expanding infrastructure in Asia and Europe, major businesses like Westport, Weichai, and Cummins propel advancements. Market growth projects 15% compound annual growth rate (CAGR) for LNG HD truck engines through 2033, fuelled by regulations and lower LNG prices [87, 88].

Performance Traits: With specific fuel consumption (SFC) reductions from improved combustion, dual-fuel modes match or surpass diesel efficiency at moderate-to-high loads. Due to excessively lean mixtures, light-load operation has historically experienced greater levels of CO and unburned HC, but modern controls like throttling or electronic metering reduce this [52]. In LNG engines, VCR technology optimizes combustion for increased efficiency by dynamically adjusting the compression ratio (CR) during operation according to load, fuel type, and circumstances. In diesel mode, VCR boosts CR (for example, to diesel-like levels) to save fuel consumption and CO_2 by up to 7.7%; in gas mode, it lowers CR to prevent knock and improve mixing, resulting in gas-mode savings of 3.1%. At partial loads—common in shipping—higher CR boosts efficiency and cuts methane slip by 30-50% via complete combustion, dropping total slip to 0.83% of gas input. Fuel efficiency rises 3-8% overall by matching CR to conditions: high CR at low loads/part-throttle maximizes expansion work; low CR at high loads avoids detonation, enabling leaner burns [89–91].

Emission Profile: Compared to pure diesel, LNG dual-fuel reduces NO_x by up to 53%, PM by over 60%, and CO_2 by 30–40% because of its reduced carbon content and cooler burning. However, along with increased HC and CO, methane slip - unburned CH4 - poses a serious GHG problem (up to 24% of vessel emissions) [92, 93].

Additionally, as Table 14 discusses, improvements have been made to LNG dual-fuel engines in recent years to increase sustainability, efficiency, and pollution reduction [94].

Table 14. Current studies on liquefied natural gas (LNG) as a replacement to diesel fuel

Author	Working Atmosphere	Conclusion	Remark
Nguyen et al. [95]	Employed a tiny amount of diesel pilot as an ignition source and LNG as the main fuel. Compared the dual fuel engine's performance to that of an equivalent diesel engine.	Dual fuel engines produced about 1780 N-m of torque at 1100 rpm, whereas diesel engines produced about 1790 N-m. Furthermore, both engines produced the same amount of power. Both engines have about identical thermal efficiency at lower rpms.	Diesel and LNG both displayed comparable power output.
Kofod and Stephenson [96]	Investigated the greenhouse gas (GHG) emissions from LNG used as long-distance truck fuel in well-to-wheel (WTT) vehicles.	LNG's WTT GHG emissions were 37.9 g CO_2e/MJ_{out}, while diesel's were 47 g CO_2e/MJ_{out}. LNG and diesel had total WTW GHG emissions of 211.7 and 262 g CO_2e, respectively.	LNG fuel emits fewer GHG than diesel fuel.
Shi et al. [97]	Conducted a life cycle assessment to ascertain the environmental emissions and energy savings of a newly manufactured diesel engine and a refurbished LNG engine.	The fuel efficiency of LNG and diesel was 26.5 m^3/100 km and 25 L/100 km, respectively, for a distance of 300,000 km.	The fuel usage of the two systems was nearly identical.

Author	Working Atmosphere	Conclusion	Remark
Selim [98]	Conducted studies on conventional diesel engines and LNG dual-fuel engines. compared the two engines' emissions and efficiency.	Dual-fuel brake specific fuel consumption rises below 45.63 kW of power. However, fuel usage decreases over this particular value. When compared to a traditional diesel engine, the dual-fuel engine's smoke density is noticeably lower.	When compared to diesel engines, the LNG engine's brake specific fuel consumption (BSFC) decreases beyond a particular speed.
Misra et al. [99]	The authors examined the NO_x production of two diesel and LNG systems using a three-way catalyst and one hydraulic hybrid diesel system.	It was discovered that diesel had the greatest NO_x emissions of all the technologies examined, whereas LNG had the slowest emissions.	LNG emits less NO_x than diesel.
Zhao et al. [100]	Examined the power and emissions of class 8 hybrid electric trucks using electricity, hydrogen, diesel, and LNG as fuels for a variety of uses.	The LNG compression ignition engine's efficiency is comparable to that of a diesel engine at part load and somewhat lower at full load.	At partial load, the efficiency of diesel and LNG is about equal.
Li et al. [101]	Created a control system for an LNG dual-fuel marine engine and contrasted several engine characteristics with those of a traditional diesel engine.	Fuel flow rates between 600 and 1800 rpm were calculated, and it was discovered that LNG consumption is less than diesel consumption at the same speed. The rate of natural gas consumption rises with rpm, however once the speed reaches 1400 rpm, it drastically drops. Additionally, the authors discovered that the cost of an LNG dual-fuel engine is less than that of a diesel engine, with a maximum reduction of 28.7% at 1300 rpm.	When it comes to fuel economy over 1400 rpm, LNG engines perform far better than traditional dual-fuel engines.

Commercial LNG engines may be able to comply with Euro VI emissions regulations by using stoichiometric operation with exhaust gas recirculation (EGR) and the three-way catalyst. The knock and durable peak in-cylinder pressure (mechanical strength) of the existing LNG engine with aluminium piston limit its thermal efficiency. Therefore, to further increase the thermal efficiency of stoichiometric LNG engines, it is crucial to develop efficient combustion and knock management procedures in higher peak in-cylinder pressure circumstances (with higher CR steel pistons) [36].

Additionally, non-saturated LNG theory-based LNG car fuel systems are being developed and tested. These aim to keep the fuel tank's liquid subcooled—that is, at higher densities and lower temperatures than saturated systems. A pressure-building circuit is used in this method. The time required to develop initial pressure after refuelling is a major problem for pressure-building circuits, which are also utilized in several saturated fuel system ideas. When viable pressure-transfer systems are unable to meet the extremely high fuel pressure needs of direct-injection natural gas engines, pumped systems have been utilized. Although there are issues with their relative cost and durability for this application, cryogenic pumps may also be utilized for lower pressure requirements. One area where technical advancement is anticipated is in fuel pressure requirements and capabilities.

In the most popular fuel system design, LNG is received and stored in the vehicle tank at a saturation pressure that is at least equivalent to the engine fuel supply pressure needed. Although this design is somewhat straightforward, fuel conditioning (saturation pressure and temperature increase) at the station is necessary [66].

For the efficient use of natural gas, the LNG engine system was equipped with a high compression ratio, manifold gas injection, and spark ignition. An LNG tank, vaporizer, LNG control valve, and gas injector were designed as parts of the LNG delivery system. To avoid BOG, extra insulating constructions were added to the LNG tank and the LNG control valves. Heat from the radiator was used to vaporize LNG. For the best fuel supply, a solenoid coil controls the gas injector. For the engine control system, a fuzzy logic computer was created.

11 Benefits to the End user for using Liquefied Natural Gas as a Fuel

The development of NGVs began in the 1930s and has continued for almost a century. After recent progress, NGV technologies are more sophisticated and user-friendly. The great performance, safety, and dependability of NGVs are the result of sophisticated electronic control and production techniques. NGVs can now travel about a thousand km, up from 50–70 km in the early years. The network of filling stations has been progressively expanded, and the size and weight of natural gas tanks have been significantly decreased. Filling NGVs is therefore just as convenient as filling petrol or diesel vehicles [3].

Whether adopting an alternative fuel is cost-effective for customers is ultimately what determines its success. Natural gas reserves in India are enormous. Natural gas is an inexpensive, low-carbon energy option for customers both domestically and internationally, and its abundance, low emissions, and dependability make it a key component of the clean energy future. Natural gas is used:

- By homes for heating and cooking;
- By industry for manufacturing essential products as varied as steel, medical equipment and fertiliser;
- By grocery stores, hotels and restaurants for heat, power and dehumidification;
- By trucks, buses and cars for clean fuel;
- By utilities and power producers to generate reliable electricity with low emissions [4].

12 Key Remarks

As the world moves toward cleaner and more sustainable energy sources, LNG has emerged as a promising transitional fuel. LNG offers significant benefits as a lower-emission substitute for coal and oil because of its better combustion properties and less carbon footprint. LNG is positioned to play a crucial role in striking a balance between present energy demands and environmental objectives, since international climate agreements like the Paris Agreement emphasise the need to reduce GHG.

LNG is natural gas cooled to -162 °C, reducing its volume by 600 times. It's ideal for storage and transport, especially where pipelines are absent. Cleaner than other fossil fuels, LNG mainly consists of methane, with impurities removed, making it suitable for industry, power, and heavy-duty transportation.

LNG benefits the automotive industry by reducing pollutants like sulphur dioxide, NO_x, and particulates, while offering quieter operation and better fuel efficiency. However, high costs of cryogenic storage and refuelling infrastructure pose challenges. Storage changes may also affect emissions and engine performance. Compared to coal and oil, LNG significantly lowers GHG, with reduced CO_2 emissions and minimal sulphur and particles. Field data shows LNG effectively decreases particle emissions, and ongoing technological advancements aim to improve engine efficiency and environmental performance further.

LNG's supply chain involves extraction, liquefaction, transportation, regasification, and distribution. It is especially practical for long-distance transport, such as to island nations and rural areas without pipelines, while pipeline transit remains more cost-effective for short distances. Various liquefaction methods—Expander-based, Mixed Refrigerant, and Cascade—offer different efficiencies and complexities. BOG, caused by heat intrusion into cryogenic tanks, is a key logistical challenge, impacting fuel management and safety. Solutions like dual-fuel engines and re-liquefaction mitigate BOG issues, and the cold energy from regasification can benefit other industries like desalination. Major exporters include the USA, Qatar, and Australia, with rising demand in Asia and Europe. Countries like India are expanding LNG infrastructure to support energy needs. Despite challenges like high costs, competition from renewables, and methane leakage, LNG remains vital in transitioning to a lower-carbon future, offering a flexible, cleaner energy source as renewable technologies advance.

13 Summary

- LNG has emerged as a strategically important transitional fuel, balancing near-term energy security with long-term decarbonisation goals. LNG use can decrease overall GHG, but methane slip remains a key challenge needing advanced technical solutions for sustainability. Using LNG to power vehicles enhances energy diversity and security. Since a significant portion of global fuel production is dedicated to automobiles, the adoption of NGVs has become a transformative development in sustainable transportation.
- LNG makes it possible to significantly reduce CO_2, SO_2, NO_2, and particle emissions as compared to coal and oil, which is in line with international climate commitments like the Paris Agreement.
- LNG is considered a cleaner fossil fuel than coal and oil, but its long-term role in the energy transition requires careful evaluation. A key challenge is managing BOG, which results from heat ingress into cryogenic storage tanks during transportation and storage. If not properly managed through reliquefication, recondensation, or direct use in engines, BOG can cause methane emissions, significantly impacting climate benefits due to methane's high global warming potential.

- Its high energy density and transport flexibility make LNG suitable for regions with limited pipeline infrastructure. Economically, LNG infrastructure (liquefaction plants, regasification terminals, storage, refuelling) demands high capital investment and long operational periods, raising concerns about infrastructure lock-in.
- LNG has advantages over diesel in the transportation industry, especially for heavy-duty and long-haul applications, in terms of increased range, operational efficiency, less noise, and lower emissions. However, the cost of the LNG vehicle is expensive mainly due to the fuel tank and NG engine; the differential is close to Rs. 12+ lakhs.
- Long-term decarbonisation solutions are represented by electric and hydrogen technologies, but their existing infrastructural and cost limitations highlight LNG's value as a bridge. LNG fuel is 20% to 30% cheaper than diesel. Compared to hydrogen, ammonia, and renewables, LNG benefits from technological maturity, higher energy density, and existing infrastructure compatibility. Its long-term viability depends on emission mitigation advances, renewable cost reductions, and scalable zero-carbon fuel options.
- Advances across the LNG value chain—including liquefaction, storage, and BOG management—have enhanced its technical and logistical viability.
- Key challenges remain, notably infrastructure development, vehicle affordability, methane slip, and cryogenic storage complexity.
- When supported by coordinated policy, technology advancement, and ecosystem-level growth, LNG can assist fuel diversification and transportation emission reduction for economies like India.

Author Contributions

Conceptualization, D.B.; Methodology, H.K.T.; Formal analysis, R.P.D.; Investigation, H.K.T.; Resources, R.P.D.; Data curation, D.B.; Writing—original draft preparation, H.K.T.; Writing—review and editing, D.B.; Visualization, R.P.D.; Project administration, A.D. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

BOG	boil-off gas
ANG	absorbed natural gas
BMEP	brake mean effective pressure
CAGR	compound annual growth rate
CCS	carbon capture and storage
CH_4	methane
CI	compressed ignition
CNG, LNG, NG	compressed natural gas, liquified natural gas, natural gas
CO_2, CO	carbon dioxide, carbon monoxide
CR	compression ratio
D100	diesel fuel
DME	dimethyl ether
EGR	exhaust gas recirculation
EXP	expander-based technology
FLNG	floating LNG
FY	Fiscal year
GAIL	Gas Authority of India Limited
GHG	greenhouse gas
H/C, C/H	hydrogen-to-carbon ratio, carbon-to-hydrogen ratio
HC, THC, NMHC	hydrocarbon, total hydrocarbon, non-methane hydrocarbon
HDV	heavy duty vehicle
HPDI	high pressure direct injection
IEEFA	Institute for Energy Economics and Financial Analysis
IFV	submerged combustion vaporiser
IMO	International Marine Organization
IWAI	Inland Waterways Authority of India
LCA	life cycle assessment
LFL	lower flammability limit
LNG FPSO	LNG floating production, storage, and off-loading
LPG	liquified petroleum gas
MMSCMD	million standard cubic meters per day
MN, SMN	methane number, service methane number

MoPNG	Ministry of Petroleum and Natural Gas
MR	mixed refrigerant technology
NGL	natural gas liquids
NGVs	natural gas vehicles
NO_x	oxides of nitrogen
ORV	open rack vaporiser
PBU	pressure build-up unit
PM, PN	particulate matter, particulate number
RCS	re-condensation system
RPT	rapid phase transition
SCM	supply chain management
SCV	submerged combustion vaporiser
SFC	specific fuel consumption
SI	spark ignition
TEG	thermoelectric generators
UFL	upper flammability limit
VCE	vapour cloud explosion
VCR	vapour compression refrigeration
VSP	vehicle specific power
WG	water-glycol
WHTC	World Harmonized Transient Cycle